TYLER MATTHEWS

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QA & Test Automation | CI/CD | Cloud Deployment | Reliability & Troubleshooting

SUMMARY

Computer Science graduate (UCF, 2026) focused on test automation, CI/CD, and reliable, observable systems. Wrote 300+ automated tests (pytest, Jest, Vitest), deployed applications to DigitalOcean with Docker, and run CI/CD through GitHub Actions. Hands-on with root-cause analysis, logging, health checks, retries, and cross-team integration troubleshooting.

EDUCATION

University of Central Florida, *Orlando, FL* | *B.S. Computer Science* | *GPA: 3.62* August 2026

Relevant Coursework: Data Structures & Algorithms, Systems Software, Machine Learning Algorithms, Computer Vision, Linear Algebra, Probability & Statistics for Engineering, Numerical Analysis

TECHNICAL SKILLS

Testing & QA: pytest, Jest, Vitest, Postman; unit, integration, regression, and API testing; coverage analysis
Cloud & DevOps: Docker, DigitalOcean Droplets, GitHub Actions CI/CD, Git/GitHub; AWS Cybersecurity course badge
Languages & Scripting: Python, Java, C, JavaScript, TypeScript, SQL, PowerShell, Bash, Linux/WSL
Backend & Data: Node.js, Express, React, REST APIs, MongoDB, MySQL, IndexedDB, pandas, NumPy, Jupyter, GeoJSON
Monitoring & Reliability: logging, health checks, job retries, error handling, root-cause analysis, troubleshooting

PROJECTS

AI Asset Generation Pipeline for Unity | *Python, pytest, GPU Workers, Unity* 2026 (In Progress)

- Building a full-stack AI pipeline for production-ready 2D game assets (textures, sprites, icons, UI) with a Python backend on local and hosted GPU workers, persistent job processing, and model management.
- Maintain 100+ automated pytest tests that catch regressions as new generation features ship.
- Implemented logging, health checks, job retries, and error handling for reliable, diagnosable GPU jobs.
- Root-caused extreme job completion times to models persisting in GPU memory and forcing CPU offloading; resolved the slowdowns with explicit model loading/offloading across the job lifecycle.
- Built Unity-native configuration, preview, and import tooling, plus authentication, storage, and deployment packaging.

ALPR Traffic Simulator | *TypeScript, React, Vitest, MapLibre, Web Workers, IndexedDB* Personal Project

- Wrote ~200 Vitest tests: ~83% line / ~91% function coverage on the core engine, ~95% on the simulation engine.
- Built a local-first browser SPA simulating synthetic traffic on real OpenStreetMap roads and recording what a hypothetical license plate reader camera network would observe.
- Ran simulation work in Web Workers, persisted data with IndexedDB, and built the map UI with MapLibre.

Polaris: Accessible Indoor Navigation | *GeoJSON, IMDF, pytest, Jest* Team Project, 2026

- Mapping lead on a 5-person team, owning the backend data tooling behind the app's indoor routing.
- Built a data-conversion pipeline producing IMDF-compliant navigation data, reducing manual processing; hand-curated the full L3Harris Engineering Center (UCF) building dataset.
- Developed tools to georeference GeoJSON building data and generate the navigation graph used by app routing.
- Integrated the IMDF bundle into the mobile app, validating compatibility with its decoder; troubleshot integration issues with mobile teammates and wrote pytest and Jest tests.

EXPERIENCE

Retail Associate | *Rocky's Ace Hardware, Jupiter, FL* 2021 - 2025

- Diagnosed customer needs and translated technical product details into clear recommendations in a high-volume store.
- Handled transactions, inventory, and competing priorities; 4 years of reliable part-time work alongside STEM coursework.